

The Exoplanet Census: Observational Bias and the Radius Gap

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1 Introduction

The cosmos is vast and has countless secrets. For centuries, our knowledge was limited to the eight planets orbiting in our solar system. This has changed over time as science and technology advanced rapidly leading to inventions of powerful telescopes which accelerated discoveries of many exoplanets orbiting stars far from our solar system. Specially missions like Kepler and TESS have led to discoveries of many exoplanets. This success has given us a huge collection of the exoplanets data.

However, the list of confirmed planets is not a true picture of the galaxy. There is a lot of missing data which reveals that our current search methods are currently limited which makes finding certain types of planets much easier than others.

The report visualizes these limitations and biases through two key data plots. The plots will focus on two major scientific features: Planets that dominate the census due to observational limits, and the Radius Gap, which defines the complexities of planetary formation and provides insights on evolution.

2 Methodology

To investigate these features, data comprising the properties of approximately 6,000 confirmed exoplanets was extracted from NASA's official exoplanet archive in the form of a csv file. Data cleaning and preparation were performed using Python libraries like Pandas and NumPy. The collection is heavily dominated by discoveries made by missions like Kepler and TESS using the Transit Method. This method of exoplanet detection relies on constant observation of a star's brightness, any minor dip or decrease in the intensity of light sourcing from an star confirms the exoplanet's availability in that particular system. Using this data, the exoplanet's mass, radius, and orbital period are calculated, details are further analysed to get more information.

2.1 Data Preparation and Technical Decisions

2.1.1 Use of Logarithmic Scale ($\log|x|$ and $\log|y|$):

The enormous range in the collected data required some steps of precise visualization. The significant variability in planetary properties such as: orbital period range from less than a day to hundreds of millions of days. To overcome this immense variability, logarithmic scale($\log|x|$ and

logly|) was used to plot visuals as using the log scale helps us to fill up and analyse a large number of content which we are not able to visualize using a simple number lined graph.

2.1.2 Technical Errors/Difficulties Faced and Lessons Learned:

During the initial data setup, a lot of errors were observed regarding the use of Python Environment like “Pandas not detected on (.venv) Kernel”, etc. During the process of plotting the graphs, a persistent error of name not found was occurring (“Error: ‘discovery_method’ not found”), it was due to the unnecessary use of an underscore(“_”) included while column naming, which was not present in the dataset’s header. This debugging process took several hours to resolve. To ensure proper use of astronomical terminologies and understanding of different detection techniques, information were gathered from the official website of National Aeronautics and Space Administration (NASA), United States. Similarly, the skills for using the markdown cell features of jupyter notebook were self-taught through websites like DataCamp. Finally, the dataset was filtered to make sure that entries with missing values are not represented, resulting in the final count of 5,999 confirmed exoplanets.

Statistical Summary of the Numerical Columns:

	discovery_year	orbital_period_days	exoplanet_rad_earth \	
count	6028.000000	5.720000e+03	6004.000000	
mean	2016.815362	7.481418e+04	5.809699	
std	4.871597	5.317788e+06	5.345519	
min	1992.000000	9.070629e-02	0.309800	
25%	2014.000000	4.341330e+00	1.820000	
50%	2016.000000	1.108732e+01	2.830000	
75%	2021.000000	3.976274e+01	11.951596	
max	2025.000000	4.020000e+08	77.342100	
	exoplanet_mass_earth	equilibrium_temp_kelvin	star_rad_solar \	
count	5997.000000	4506.000000	5737.000000	
mean	388.935858	916.354672	1.508969	
std	1111.770520	465.137431	3.956330	
min	0.020000	34.000000	0.011500	
25%	4.130000	569.000000	0.772344	
50%	9.100000	823.000000	0.953000	
75%	182.433506	1166.750000	1.250000	
max	12651.500000	4050.000000	88.475000	
	star_mass_solar	star_metallicity	star_temp_kelvin	star_luminosity \
count	6021.000000	5505.000000	5759.000000	5743.000000
mean	0.941696	0.015257	5410.571891	-0.135212
std	0.421318	0.188272	1753.619656	0.777062
min	0.009400	-1.000000	415.000000	-6.090000
25%	0.777000	-0.080000	4917.500000	-0.464655
50%	0.940000	0.020000	5554.000000	-0.071600
75%	1.090000	0.130000	5902.000000	0.313145
max	10.940000	0.600000	57000.000000	3.800000

	star_log_gravity	star_age_gyr	star_rot_vel	star_rot_period_days \
count	5732.000000	4757.000000	2210.000000	828.000000
mean	4.383067	4.364686	5.591013	52.300226
std	0.440383	2.987781	14.051302	317.835947
min	0.541000	0.000000	0.000000	-35.732000
25%	4.293000	2.330000	1.700000	12.454000
50%	4.453000	3.980000	2.500000	23.610000
75%	4.577000	5.620000	4.500000	40.100000
max	8.07000	16.100000	299.000000	7900.000000

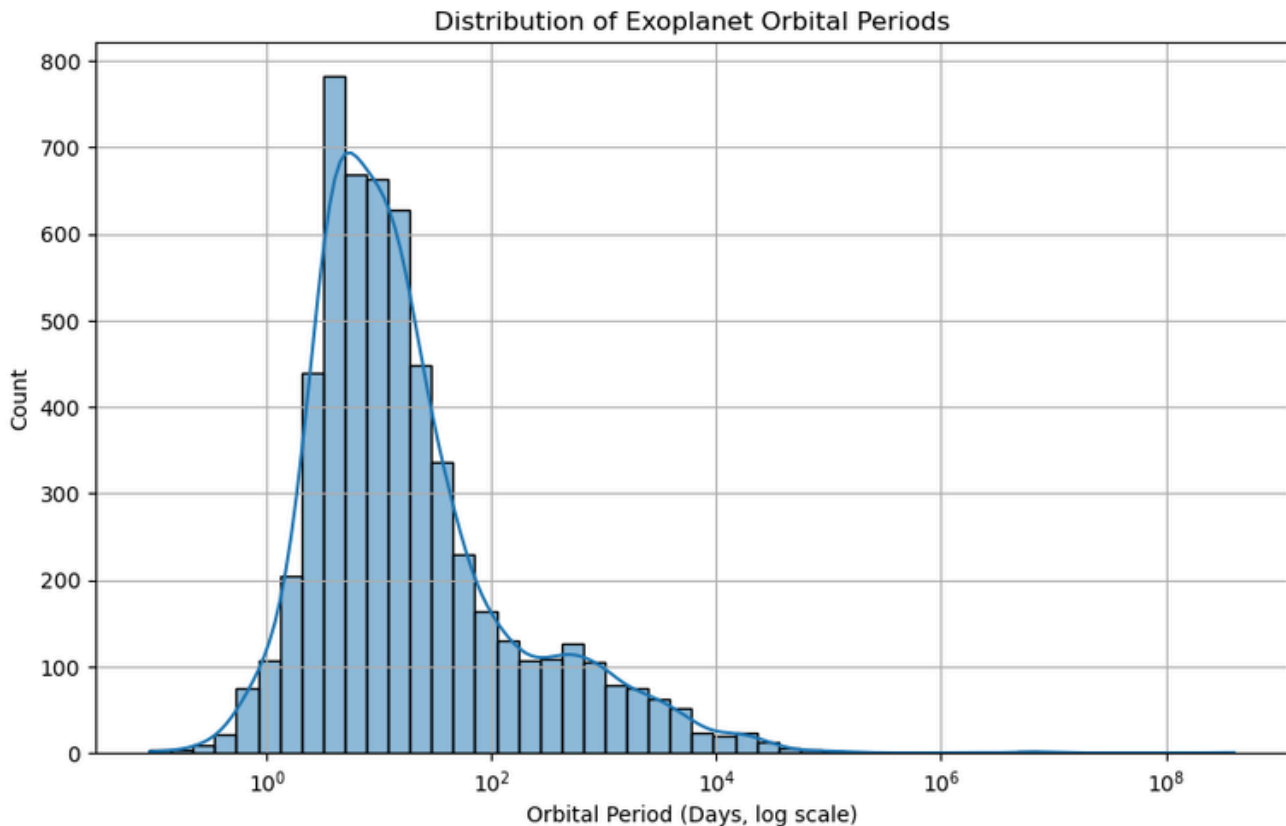
	star_rad_vel	star_ra_degrees	star_dec_degrees	system_dis_parsec \
count	2333.000000	6028.000000	6028.000000	6002.000000
mean	-1.575847	233.646564	18.619407	703.090496
std	32.836335	91.340697	36.665238	1175.439783
min	-121.900000	0.185606	-88.121111	1.301190
25%	-20.208390	175.097224	-10.253125	105.505000
50%	-1.388230	285.132349	39.294234	385.231000
75%	16.905000	293.281704	45.426491	838.955750
max	244.987150	359.974984	86.860343	8500.000000

	system_visual_magnitude	system_infrared_magnitude
count	5757.000000	5766.000000
mean	12.567574	10.412454
std	3.064943	2.998351
min	0.872000	-3.044000
25%	10.675000	8.402000
50%	13.277000	11.182000
75%	14.952000	12.756750
max	44.610000	33.110000

3 Descriptive Statistics

The statistical summary of the raw data reveals that exoplanet properties are vastly widespread, while those of the host stars are compact. For example, the orbital period days for the exoplanets range from a minimum of 9.07×10^{-2} days (less than 2.2 hours) to a maximum of over 400 million days, indicating enormous variability. This extreme variance in the standard deviation which is approximately 5.3×10^6 , confirms the need for a logarithmic approach in our visualizations. On the other hand, the host stars mass and radius are typically close to the solar mass, clustering tightly around $0.94M_{\odot}$. This is a key finding as it suggests that our current detection techniques are limited for finding planets that orbit predictable, Sun-like star

Plotting a Histogram to compare the Orbital Periods of all Exoplanets:

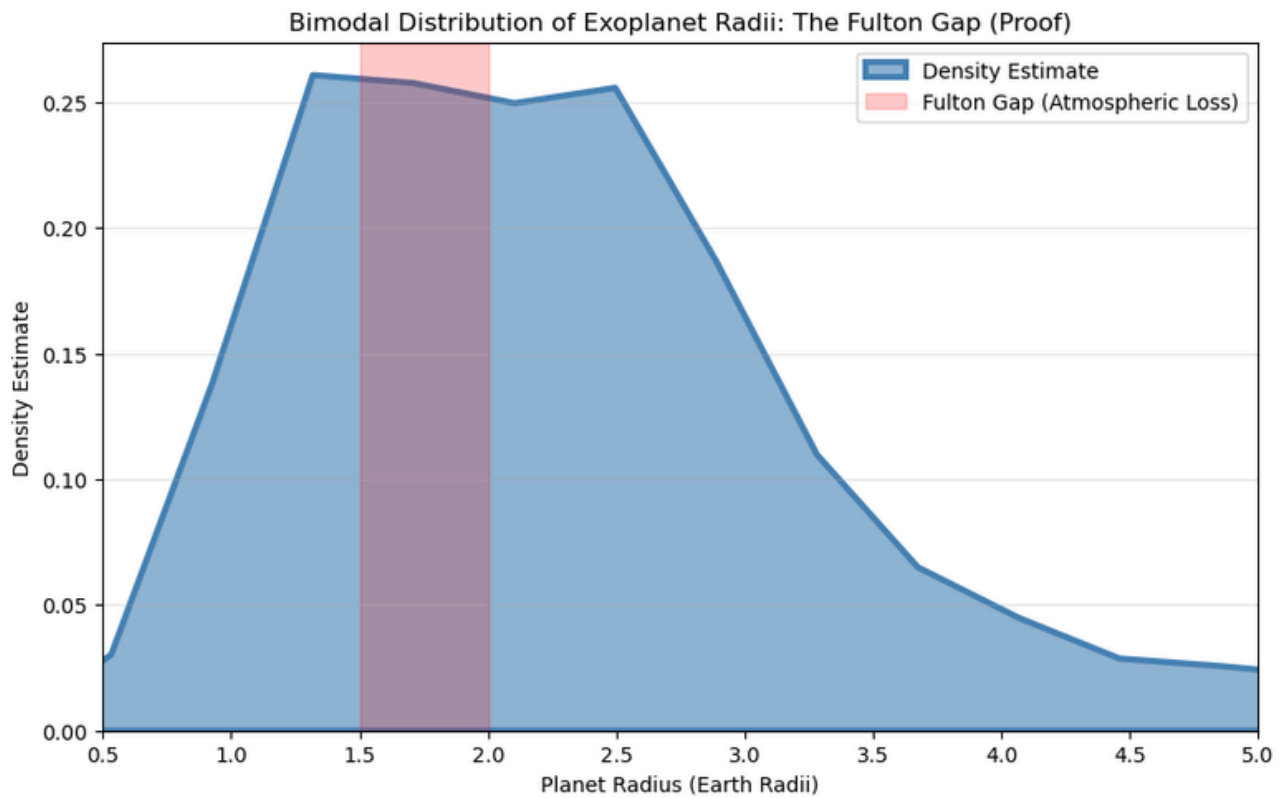


4 Distribution of Exoplanets with relation to their Orbital Periods

The histogram above displays the frequency distribution of exoplanets with regard to their orbital periods.

The analysis is performed on a logarithmic scale to properly visualize the extreme range of orbital times, from planets circling their stars in less than a day to those taking several years. The 100 represents that the planets complete their rotation around the host star in a day (Earth's). The data reveals that most of the exoplanets have an orbital period of 30-60 days, which concludes that most of the exoplanets are very close to their host stars. This is also because the current detection techniques favour finding planets that are relatively close to their host stars. In the far right, you can see a very tiny bulge reflecting a few number of exoplanets with very large orbital periods.

In conclusion, the strange distribution of the exoplanets confirms that there is a strong observational bias in current detection techniques. With most of the planets falling in that short-orbital period planets category, it is confirmed that the census is currently limited, making the number of longperiod planets extremely rare and valuable.



5 References

The identification of the Radius Gap in this analysis was guided by the California-Kepler Survey (CKS) III, 2017. For the physical foundation of this phenomenon, I referred to the work of Fulton and Petigura (2017), specifically focusing on the following sections:

Section 4: The Planet Radius Gap

Section 5.3.1: Photoevaporation

Section 5.3.3: Impact Erosion

Section 5.4: Core Mass Distribution

6 The Bimodal Distribution: Visualizing the Radius Gap

The KDE plot above illustrates the frequency of exoplanets relative to their physical size (Earth radii).

The data reveals two distinct peaks: one centred around $1.3 R_{\oplus}$ (rocky Super-Earths) and another near $2.4 R_{\oplus}$ (gaseous Sub-Neptunes). The red shaded region highlights the Fulton Gap, a significant “plunge” in the population where planets are notably missing. In my analysis, this descent is even more distinct, extending toward the $2.1\text{--}2.2 R_{\oplus}$ range before recovering.

This “valley” serves as clear evidence of the process of atmospheric disappearance. It suggests that planets in this size range cannot maintain their atmosphere; they either lose their atmospheric gas entirely and “shrink” into the first peak or are massive enough to hold onto them and stay in the 5 second. Simply, planets don’t just gradually turn from rocks into gas giants; instead, intense heat and radiation from their stars force a sudden transformation.

As observed in the bimodal distribution of the KDE plot, there is a distinct deficit of planets between 1.5 and $2.0 R_{\oplus}$. This is famously known as the Radius Gap, and it is driven by two primary physical phenomena:

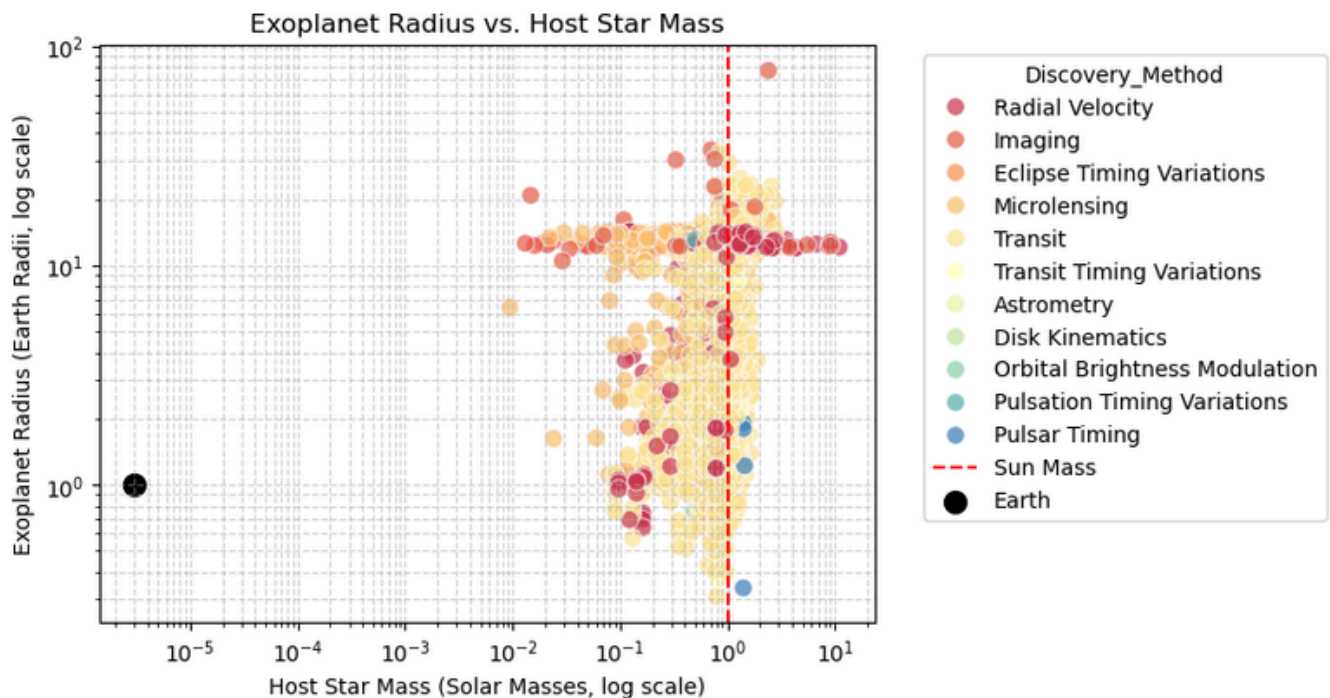
1. Photoevaporation:

High-energy radiation from the host star heats the planetary atmosphere, causing gas to escape.

2. Impact Erosion:

Occurs when foreign bodies like comets, meteoroids, satellites or neighbouring planets impact the planet.

Plotting 5999 Exoplanets with column names to collect precise evidences:



7 Discovery Method Distribution

The graph above compares the exoplanet radius and the Stellar mass of its system. Different colours of dots in the graph represent the discovery method of these exoplanets.

The measurement is taken in logarithmic scale because of the significant variation in planetary radii and Stellar masses. The y-axis represents the exoplanet radii in the multiples of Earth's radius. For example the data at 101 means that the planets radius is 10 times than that of Earth. The x-axis is identical to the y-axis with the data representing Stellar mass in respect to that of the Sun.

The resulting plot illustrates the properties of 5,999 confirmed exoplanets. While the data includes various techniques like Radial Velocity, Imaging, Microlensing, Pulsar Timing, etc, the majority 6 of the dataset is dominated by the Transit method (represented by the orange-yellow marks), specifically from Kepler and TESS missions. Information regarding this detection technique and its biases was sourced from official NASA Science Documentation.

As observed, the points (representing exoplanets) are widespread in the right side of the graph. It suggests that most of the exoplanets host star have around the same mass as the Sun in our solar system does as most of them fall between the 10^{-1} to 10^1 . Whereas, there is a widespread of distribution of data along the y-axis proving the variation in exoplanets radii as we discussed earlier.

8 Conclusion

In conclusion, the current exoplanet census is defined by two factors: limitation of our detection methodology and the physics of planets formation itself.

The analysis of orbital period distributions clearly showcases our observational limitations, with Transit method being the leading method of discovery. This creates a natural bias, making the discovery of short-period, gassy planets (known as Hot Jupiters) much easier. This bias shows humanity still hasn't mastered the art of finding exoplanets, especially planets those are habitable. Furthermore, the comparison between exoplanet radius and Stellar mass highlighted a Radius Gap among the discovered planets. The gap is the result of a evolutionary process like photoevaporation which changes the planet's geography over time.

Ultimately, by understanding the observational bias and the physical constraints implied by the featured data, we must look forward to directing future missions toward correcting these biases.